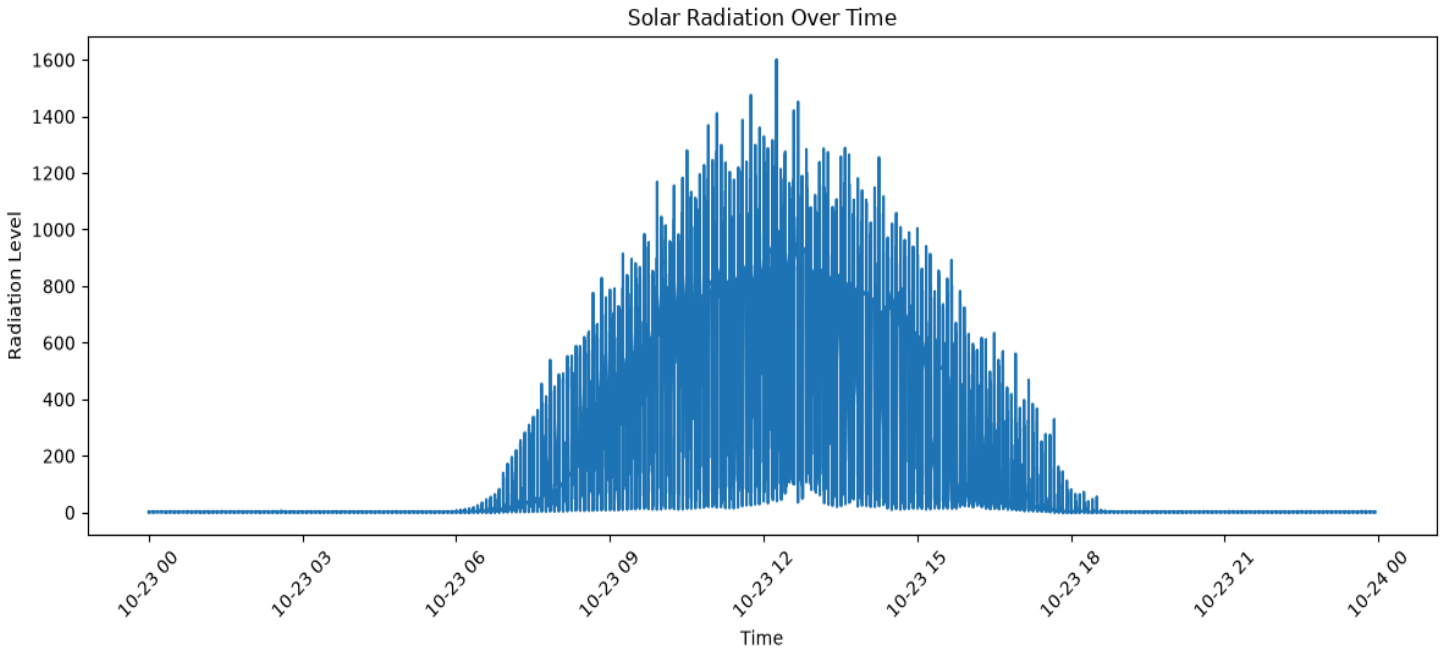
Part A-DATA VISUALIZATION

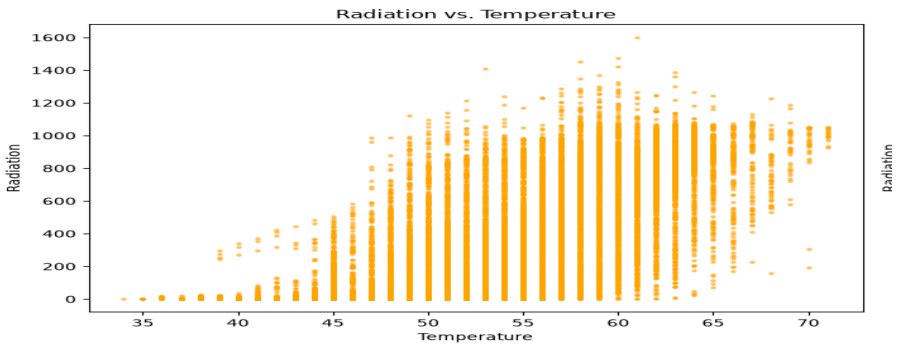
Plot: Time vs Radiation
Insights:

- **The Solar Bell Curve:** Radiation follows a strict, predictable bell curve that begins exactly at sunrise, peaks sharply between 12:00 PM and 2:00 PM, and drops to zero at sunset.
- **The Danger Zone:** Solar flares (the extreme spikes) are not distributed randomly throughout the 24-hour cycle. They almost exclusively hide within the natural midday peak, making solar noon the highest-risk period for grid overloads.



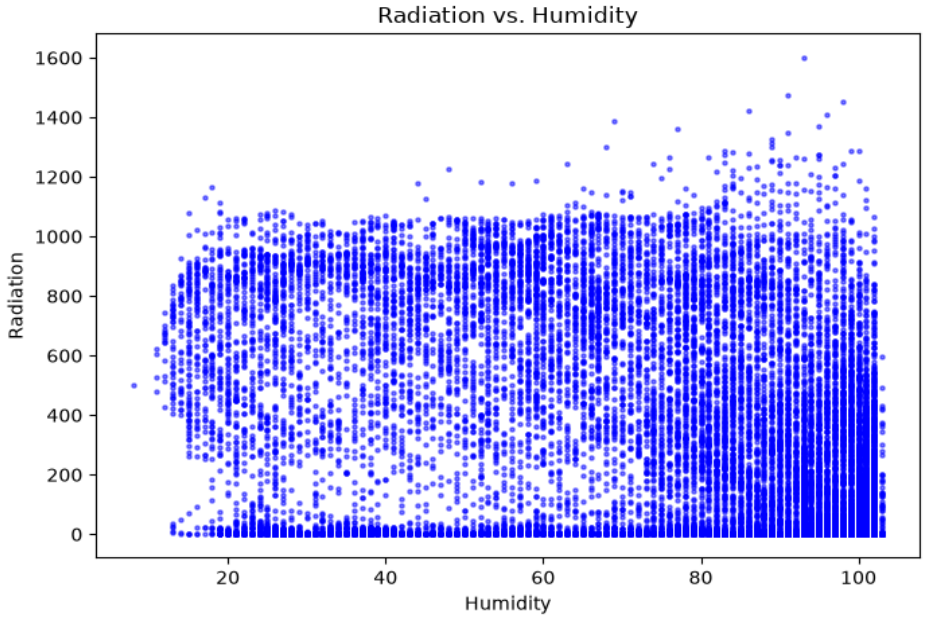
Plots: Temperature vs Radiation
Insights:

- **The Thermal Engine (Temperature):** There is a strict, positive correlation between temperature and radiation. As expected, the higher the radiation, the higher the surface temperature. The data tightly clusters together, proving temperature is the most reliable secondary indicator of solar activity.



Plots: Humidity vs Radiation
Insights:

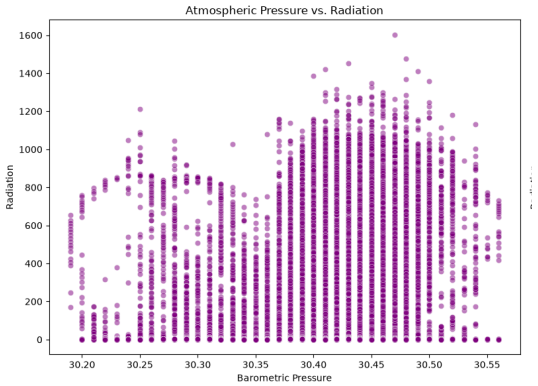
- **Moisture Paradox (Humidity):** The data reveals a surprising hidden insight. Common logic suggests that high humidity (thick, cloudy, moisture-heavy air) would block the sun and reduce radiation. However, the scatter plot shows massive radiation spikes occurring even at high humidity levels. The absolute highest flare on record (1601.26) struck when humidity was at a suffocating 93%. Thick atmosphere does not shield the base from magnetic solar flares.



Plots:Atmospheric Pressure **vs.** Radiation
Insights:

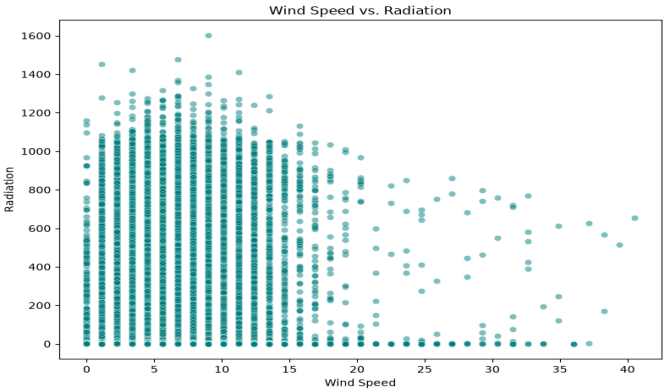
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- The purple scatter plot reveals that the absolute highest solar flares (radiation exceeding 1400) do not happen randomly; they cluster tightly within a highly specific barometric pressure window of 30.40 to 30.50.When the pressure drops below 30.35 or rises above 30.50, peak radiation levels taper off significantly, rarely breaking 1200.
- Extreme solar flares require highly stable, specific high-pressure atmospheric conditions to reach the surface at maximum intensity.



Plots:Wind Speed **vs.** Radiation
Insights:

- The teal scatter plot demonstrates a clear ceiling effect caused by wind. Every single massive radiation spike (above 1400) occurred when wind speeds were calm to moderate, strictly remaining under 15.
- As wind speeds increase past 20, radiation levels are heavily suppressed, struggling to even break past 800.
- High winds are likely indicators of heavy storm fronts or atmospheric turbulence that physically scatter or block intense solar radiation. A windy day acts as a natural shield for the base's grid.



Part B- DRAWING INFERENCES

1. Peak Solar Activity Window: reaches its absolute highest levels during solar noon, specifically between **12:00 PM and 2:00 PM**
2. The Primary Triggers (Temperature & Hour): The model mathematically proved that **Temperature (58.1%)** and the **Hour of the day (29.2%)** are the two most critical features for predicting solar activity. Together, they control over 87% of the radiation output. The base should rely almost exclusively on thermometers and clocks to predict incoming surges.
3. The Irrelevant Factors: Wind Speed (2.4%), Atmospheric Pressure (3.7%), and Humidity (6.6%) are practically irrelevant. You can confidently infer that intense wind or high moisture in the air will not stop a solar flare from hitting the grid.
4. The Humidity Paradox **:**While temperature and radiation rise together predictably, humidity breaks conventional expectations. Common logic dictates that heavy moisture in the air (cloud cover) should block sunlight, but the scatter plots show extreme radiation spikes penetrating even thick atmospheres. The absolute highest radiation flare on record (1601.26) struck when humidity was at a suffocating 93%.
5. Statistical outliers—defined as radiation exceeding the 1351.12 safety threshold and peaking at a maximum of 1601.26—represent isolated solar magnetic flares that are highly concentrated in the volatile month of October. The data proves these intense bursts easily bypass dense atmospheric moisture, striking even at 93% humidity, but successfully penetrate the surface only during periods of extreme atmospheric stillness, strictly clustering when wind speeds remain under 15 and barometric pressure stabilizes precisely between 30.40 and 30.50.

   Reasons for such solar flares can be:
   - **High-Energy Penetration:** Solar flares emit extreme UV and X-ray wavelengths. Unlike visible light, these high-energy waves do not scatter against water droplets, allowing them to punch directly through dense atmospheric moisture even at 93% humidity.

- **Atmospheric Lensing (Stillness):** Stable barometric high pressure (30.40–30.50) combined with low wind creates a sinking, compressed air mass. This eliminates upper-level turbulence and dust, acting as a transparent magnifying glass that delivers solar radiation to the surface at absolute maximum intensity.
- **Equinox Vulnerability:** The frequency of October anomalies aligns with the Russell-McPherron effect. During the post-equinox weeks, Earth's axial tilt aligns its magnetic poles with the sun's magnetic field, temporarily cracking the magnetosphere and allowing solar bursts to slip through Earth's natural shields.

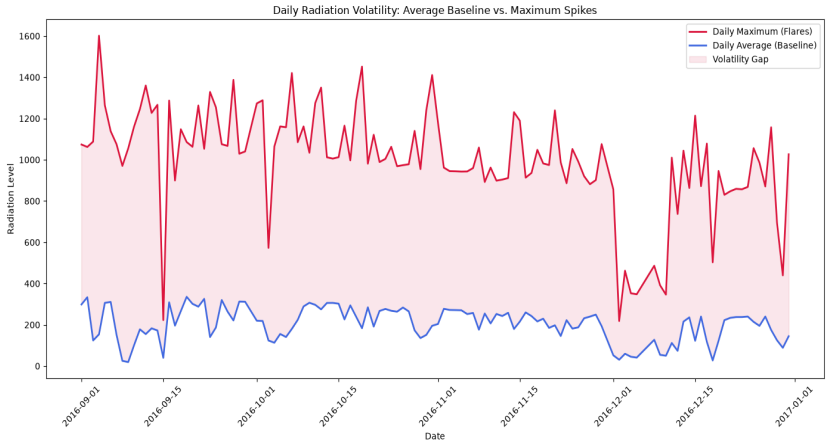
PART C- SOLAR DAY ANALYSIS

★ TOP 5 DAYS (HIGHEST AVERAGE RADIATION)

Date_Only	Max_Radiation	Avg_Radiation
2016-09-19	1085.75	335.50
2016-09-02	1062.01	333.47
2016-09-22	1053.17	325.52
2016-09-25	1075.29	319.92
2016-09-28	1029.30	312.40

★ Top 5 Days (by Maximum Radiation):

Date	Max_Radiation	Avg_Radiation
09/04/2016 12:00:00 AM	1601.26	153.584321
10/19/2016 12:00:00 AM	1451.41	183.265208
10/07/2016 12:00:00 AM	1420.39	180.155599
10/31/2016 12:00:00 AM	1410.52	194.996829
09/27/2016 12:00:00 AM	1387.17	220.851210



1. MAXIMUM RADIATION EVENT

Date of Highest Radiation: 2016-09-04
Maximum Radiation Recorded: 1601.26 units

2. ENVIRONMENTAL CONDITIONS DURING PEAK

Time of Peak: 2026-10-23 12:15:04
Temperature: 61°
Humidity: 93%
Wind Speed: 9.0

3. SUNRISE COMPARISON
Sunrise Time: 06:08:00
Time Elapsed Since Sunrise: 6 hours and 7 minutes

SOLAR RADIATION PREDICTION

RMSE (Average Error Margin): 95.82

R-squared Score: 0.91

IMPORTANT FEATURES AND THEIR ROLE

Temperature: 58.1%
Humidity: 6.6%
Hour: 29.2%
Speed: 2.4%
Pressure: 3.7%

EXTRA OBSERVATIONS

The Wind Paradox: Stagnation vs. Clear Skies

- The Data: Perfectly calm air (0–5 speed) is required to capture extreme 1600-unit flares, but it suppresses the normal daily average to just 457 units. A moderate breeze (10–15 speed) boosts the daily average by 24% to 569 units.
- The Reason: Zero wind causes atmospheric stagnation, allowing low-level dust and moisture to hang in the air and block standard sunlight. A moderate breeze physically clears this haze, keeping the sky transparent for a higher everyday power yield.

Seasonal Volatility: September Stability vs. October Extremes

- The Data: September provides the most consistent baseline, containing all top 5 days for highest average radiation (312+ units). October sees a sharp drop in daily averages but a massive increase in extreme maximum spikes (1400+ units).
- The Reason: September offers reliable, day-long solar generation. As the season shifts into October, the atmosphere becomes highly erratic, forcing the grid to rely on capturing short, intense midday flares rather than steady, day-long sunlight.